

Exercise 05

Relational algebra and functional dependencies

Given the following relations:

Students:	SName	SMatrikel	SBirthday
	Heintje	2143	05.04.2001
	Eva	3333	06.04.2001
	Luise	3334	06.04.2002
	Daniel	3335	07.04.2002
	Daniel	3336	07.04.2001
	Heintje	3337	17.04.2001

Lecturers:	LName	LOffice	LTel
	Klaus	C201	123
	Maria	D22	null
	Marlene	C204	456
	Matze	E4	null

Event:	ENAME	ESemester	ERoom	ELecturer
	Beachvolleyball	ss17	Beach	Maria
	Beachvolleyball	ss18	Beach	Maria
	Hanggliding	ss17	Beach	Maria
	Hanggliding	ss18	Beach	Maria
	Sackhüpfen	ws17	null	Klaus
	Sackhüpfen	ws18	null	Klaus
	Dance Gymnastics	ss17	D111	Klaus
	Dance Gymnastics	ss18	D111	Klaus

Task 1: Anomalies

Specify a deletion anomaly, insertion anomaly and update anomaly for the relation `Lecturers`.

deletion anomaly: deleting a lecturer leads to losing their office room and telephone number

insertion anomaly: inserting a new lecturer without an office

update anomaly: updating a lecturer's without updating Event

Task 2: Keys, superkeys and foreign keys

Justify the following statements and show them formally by applying the definitions:

- a) *SName* in *Students* cannot be a key.
- b) *LName* from *Lecturers* does not meet the foreign key constraint for *ELecturer* in *Events*.
- c) *ELecturer* in *Events* fulfils the foreign key constraint for *LName* from *Lecturers*.
- d) *ESemester*, *ERoom*, *ELecturer* in *Events* fulfils not the superkey property in the given instance of the database.
- e) *LName*, *LOffice* fulfils the superkey property of *Lecturers* in the given instance of the database.
- f) *LOffice* in *Lecturers* fulfils the key constraint.
- g) The relational schema *R* is a superkey of *R*.
- h) *ENAME*, *ESemester* fulfils the superkey constraint of *Events*.

Task 3*: Functional dependencies, Splitting Cover and Key Heuristic

Starting from our example using *Lecturers*, *Students*, *Events* and *Grades* we now assume that there are only the following attributes: *SName*, *SMatr*, *SBDay*, *LName*, *LOffice*, *LTel*, *ENAME*, *ESemester*, *ERoom* and *Grade* stored in one relation *R* (yes, that's bad!).

We assume, that the following functional dependencies *FD* are valid:

$$\begin{aligned} SMatr &\rightarrow SMatr, SName, SBDay \\ LName &\rightarrow LOffice, LTel \\ EName, ESemester &\rightarrow ERoom \\ SMatr, EName, ESemester &\rightarrow Grade \\ SMatr &\rightarrow SName \\ LName, LOffice &\rightarrow LTel \\ SMatr, EName, ESemester &\rightarrow ERoom \end{aligned}$$

- a) Simplify this set *FD* using the SPLITTING algorithm.
- b) Provide 10 *FDs* that are in $\{FD\}^+$ for the functional dependencies *FD* that are not in the list above.
- c) What is $\{SMatr\}^+_{FD}$?
- d) What is $\{SMatr, LName\}^+_{FD}$?
- e) What is $\{SName, LOffice, ERoom\}^+_{FD}$?
- f) Simplify the set from part (a) of the task further using the COVER algorithm. What can you see, when you compare the *FDs* to the relations of Task 1?
- g) Determine the keys using Key Heuristic.